T0 Quantum Field Theory: Complete Extension QFT, Quantum Mechanics and Quantum Computers in the T0-Framework From fundamental equations to technological applications

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Abstract

This presentation outlines how central aspects of quantum field theory, quantum mechanics, and quantum computer technology can be formulated within the T0-Framework. Based on the time-mass duality $T_{\text{field}} \cdot E(x,t) = 1$ and the universal parameter $\xi = \frac{4}{3} \times 10^{-4}$, extensions of the Schrödinger and Dirac equations, corrections to the Bell inequalities, and concepts for deterministic quantum computers are proposed. The document presents a deterministic, local-realistic interpretation that offers a route to solving the measurement problem of quantum mechanics, and it outlines possible applications in quantum technology. Many of these approaches are not yet derived in detail in this early document and are therefore marked as speculative.

Note on versions. The German version of this document is more extensive than this English one. Earlier revisions shortened or changed parts of the English version. Where the two differ, the more extensive German version applies.

Status markers. Statements marked **[S]** are *speculative*: a hint or conjecture that has not yet been derived or numerically checked in the corpus.

.1 Introduction: T0 Approach to QFT and QM

The T0-Theory proposes extensions of quantum field theory and of the basic equations of quantum mechanics, and it opens new approaches for quantum computer technologies.

T0 Basic Principles for QFT and QM

Fundamental T0 Relations:

$$T_{\text{field}}(x,t) \cdot E(x,t)(x,t) = 1 \quad \text{(Time-Energy Duality)} \tag{1}$$

$$\Box \delta E + \xi \cdot \mathscr{F}[\delta E] = 0 \quad \text{(Universal Field Equation)} \tag{2}$$

$$\mathscr{L} = \frac{\xi}{E_{\text{Pl}}^2} (\partial \delta E)^2 \quad \text{(T0 Lagrangian Density)} \tag{3}$$

.2 T0 Field Quantization

Canonical Quantization with Dynamic Time

The central step of T0-QFT lies in the treatment of time as a dynamic field:

T0 Canonical Quantization

Modified Canonical Commutation Relations:

$$[\hat{\phi}(x), \hat{\pi}(y)] = i\hbar\delta^3(x-y) \cdot T_{\text{field}}(x,t) \tag{4}$$

$$[E(\hat{x},t)(x), \hat{\Pi}_E(y)] = i\hbar\delta^3(x-y) \cdot \frac{\xi}{E_{\text{Pl}}^2} \tag{5}$$

The field operators take an extended form:

$$\hat{\phi}(x,t) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_k \cdot T_{\text{field}}(t)}} \left[\hat{a}_k e^{-ik\cdot x} + \hat{b}_k^\dagger e^{ik\cdot x}\right] \tag{6}$$

T0-Modified Dispersion Relation

The energy-momentum relation is modified by the time field:

$$\boxed{\omega_k = \sqrt{k^2 + m^2} \cdot \left(1 + \xi \cdot \frac{\langle \delta E \rangle}{E_{\text{Pl}}}\right)} \tag{7}$$

.3 T0 Renormalization: Natural Cutoff

T0 Renormalization

Natural UV-Cutoff:

$$\Lambda_{\text{T0}} = \frac{E_{\text{Pl}}}{\xi} \approx 9.2 \times 10^{22} \text{ GeV} \tag{8}$$

At this scale the loop integrals are naturally cut off; a detailed proof of convergence is not given here.

The beta functions are modified by T0 corrections:

$$\beta_g^{\text{T0}} = \beta_g^{\text{SM}} + \xi \cdot \frac{g^3}{(4\pi)^2} \cdot f_{\text{T0}}(g) \tag{9}$$

.4 T0 Quantum Mechanics: Basic Equations in the T0 Framework

T0-Modified Schrödinger Equation

The Schrödinger equation receives an extension through the dynamic time field:

T0 Schrödinger Equation

Time Field-Dependent Schrödinger Equation:

$$i\hbar \cdot \frac{T_{\text{field}}(x,t)}{T_0} \frac{\partial \psi}{\partial t} = \hat{H}_0 \psi + \hat{V}_{\text{T0}}(x,t)\psi \tag{10}$$

where:

$$\hat{H}_0 = -\frac{\hbar^2}{2m}\nabla^2 + V_{\text{extern}}(x) \tag{11}$$

$$\hat{V}_{\text{T0}}(x,t) = \xi \cdot \frac{\delta E(x,t)^2}{E_{\text{Pl}}} \tag{12}$$

The time field stands in front of the time derivative as the dimensionless ratio T/T_0 (A142); with a dimensionful $T = 1/m$ the equation would be inhomogeneous, because the time-field term would then carry a different energy dimension than $\hat{H}\psi$. For a state of energy E the phase runs with ET_0/T, i.e. at the local clock rate, and the amplitude follows the local energy density of the internal configuration of the state, not a probability (Doc. 230). This is the gravitational redshift (Doc. 078): gravitation is already contained in the equation; beyond general relativity this form predicts nothing. No additional field and no graviton is needed for this: $T = 1/m$ is the dual reading of the mass, and gravitation shows itself as an effect in the redshift, read as curvature of space, as a change of mass (Doc. 078) or as fractal path lengthening (Doc. 041).

For a mass in superposition this means, in this reading: the individual outcome is fixed in the interaction with the measuring arrangement (Doc. 230), and the gravitational field follows the realised position. The mass is not distributed as an average over two positions, as semiclassical gravity assumes with the expectation value of the energy-momentum tensor as its source [1, 2]. This is compatible with the experiment of Page and Geilker, in which the gravitational field followed the respectively realised position of the masses and not the average [3]. How a mass acts gravitationally before any interaction has not been derived so far; this question is addressed by the proposed experiments on gravitationally induced entanglement [4, 5].

Physical Interpretation

The T0 modification has three consequences:

1. **Variable Time Evolution:** The quantum evolution proceeds more slowly in regions of high energy density
2. **Energy Field Coupling:** The T0 potential couples quantum particles to local field fluctuations
3. **No Derived Deviation beyond the Redshift:** Beyond the gravitational redshift, the theory currently predicts no measurable deviations from standard QM predictions; this does not rule such deviations out. A ξ effect could not be resolved with present precision

Hydrogen Atom with T0 Corrections

For the hydrogen atom, the result is:

$$E_n^{\text{T0}} = E_n^{\text{Bohr}} \left(1 + \xi \frac{E_n}{E_{\text{Pl}}}\right) \tag{13}$$

$$= -13.6 \text{ eV} \cdot \frac{1}{n^2} \left(1 + \xi \frac{13.6 \text{ eV}}{1.22 \times 10^{19} \text{ GeV}}\right) \tag{14}$$

The correction is tiny (relative 1.5×10^{-31}, absolute $\sim 2 \times 10^{-30}$ eV) and lies far beyond present spectroscopic resolution; it is measurable only in principle.

T0-Modified Dirac Equation

Relativistic quantum mechanics is extended by the T0 time field:

T0 Dirac Equation

Time Field-Dependent Dirac Equation:

$$\left[i\gamma^\mu \left(\partial_\mu + \frac{\xi}{E_{\text{Pl}}} \Gamma_\mu^{(T)}\right) - m\right] \psi = 0 \tag{15}$$

where the T0 spinor connection is:

$$\Gamma_\mu^{(T)} = \frac{1}{T(x,t)(x)} \partial_\mu T(x,t)(x) = -\frac{\partial_\mu \delta E}{\delta E} = -\partial_\mu \ln \delta E \tag{16}$$

Spin and T0 Fields

The spin properties are modified by the time field:

$$\vec{S}^{\text{T0}} = \vec{S}^{\text{Standard}} \left(1 + \xi \frac{\langle \delta E \rangle}{E_{\text{Pl}}}\right) \tag{17}$$

$$g_{\text{factor}}^{\text{T0}} = 2 + \xi \frac{m^2}{M_{\text{Pl}}^2} \tag{18}$$

[S] A contribution of this term to the anomalous magnetic moments of the electron and muon is not shown by this; because of the factor m^2/M_{Pl}^2 it is very small for both particles.

.5 T0 Quantum Computers: Approaches to Information Processing

Deterministic Quantum Logic

The T0 theory suggests a different concept of quantum computers **[S]**:

T0 Quantum Computer Principles

Fundamental Differences from Standard QC:

- **Deterministic Evolution:** Quantum gates are predictable in principle
- **Energy Field-Based Qubits:** $|0\rangle$, $|1\rangle$ as energy field configurations
- **Time Field Control:** Manipulation through local time field modulation
- **Natural Error Correction:** Self-stabilizing energy fields

T0 Qubit Representation

A T0 qubit is realized through energy field configurations:

$$|0\rangle_{T0} \leftrightarrow \delta E_0(x,t) = E_0 \cdot f_0(x,t) \tag{19}$$

$$|1\rangle_{T0} \leftrightarrow \delta E_1(x,t) = E_1 \cdot f_1(x,t) \tag{20}$$

$$|\psi\rangle_{T0} = \alpha|0\rangle + \beta|1\rangle \leftrightarrow \alpha\delta E_0 + \beta\delta E_1 \tag{21}$$

T0 Quantum Gates

Quantum gates are realized through targeted time field manipulation: **T0 Hadamard Gate:**

$$H_{T0} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \cdot \left(1 + \xi\frac{\langle\delta E\rangle}{E_{Pl}}\right) \tag{22}$$

T0 CNOT Gate:

$$\text{CNOT}_{T0} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \cdot \left(\mathbb{I} + \xi\frac{\delta E(x,t)}{E_{Pl}}\sigma_z \otimes \sigma_x\right) \tag{23}$$

Quantum Algorithms with T0 Improvements

T0 Shor Algorithm

The factorization algorithm would be modified by the deterministic T0 evolution **[S]**:

$$P_{Erfolg}^{T0} = P_{Erfolg}^{Standard} \cdot \left(1 + \xi\sqrt{n}\right) \tag{24}$$

where n is the bit length of the number to be factored. For RSA-2048, this estimate would give a success probability improved by $\sim 10^{-2}$.

T0 Grover Algorithm

The database search would be modified by energy field focusing **[S]**:

$$N_{Iterationen}^{T0} = \frac{\pi}{4}\sqrt{N}(1 - \xi \ln N) \tag{25}$$

This would give a small logarithmic reduction of the iteration count for large databases.

.6 Bell Inequalities and T0 Locality

T0-Modified Bell Inequalities

The Bell inequalities receive small corrections through the T0 time field:

T0 Bell Corrections

Modified CHSH Inequality:

$$|E(a,b) - E(a,b') + E(a',b) + E(a',b')| \leq 2 + \xi\Delta_{T0} \tag{26}$$

where Δ_{T0} is the time field correction:

$$\Delta_{T0} = \frac{\langle|\delta E_A - \delta E_B|\rangle}{E_{Pl}} \tag{27}$$

Local Reality with T0 Fields

The T0 theory proposes a local-realistic interpretation of quantum correlations **[S]**:

Hidden Variable: The Time Field

The T0 time field acts as a local hidden variable:

$$P(A,B|a,b,\lambda_{T0}) = P_A(A|a,T_{field,A}) \cdot P_B(B|b,T_{field,B}) \tag{28}$$

where $\lambda_{T0} = \{T_{field,A}(t), T_{field,B}(t)\}$ are the local time field configurations.

Superdeterminism through T0 Correlations

The T0 time field leads to a superdeterministic interpretation without "spooky action at a distance":

$$T_{field,A}(t) = T_{field,common}(t - r/c) + \delta T_{field,A}(t) \tag{29}$$

$$T_{field,B}(t) = T_{field,common}(t - r/c) + \delta T_{field,B}(t) \tag{30}$$

The common time field history is meant to explain the correlations without violating locality. This requires that source and measurement settings are statistically dependent through this common history (superdeterminism); without that assumption, the Bell bound holds for the factorized form above.

.7 Experimental Tests of T0 Quantum Mechanics

High-Precision Interferometry

Atom Interferometer with T0 Signatures

A possible T0 phase shift in atom interferometers would take the form **[S]**:

$$\Delta\phi_{T0} = \frac{m \cdot v \cdot L}{\hbar} \cdot \xi\frac{\langle\delta E\rangle}{E_{Pl}} \tag{31}$$

For cesium atoms ($m = 2.21 \times 10^{-25}$ kg) in a 1-meter interferometer with $v = 1$–10 m/s:

$$\Delta\phi_{\text{T0}} \sim 2 \times 10^{-23} \ldots 2 \times 10^{-22} \text{ rad} \times \frac{\langle \delta E \rangle}{1 \text{ eV}} \tag{32}$$

This lies far below the present resolution of atom interferometers; a measurement of this term is currently not possible.

Gravitational Wave Interferometry

A possible T0 correction in gravitational wave signals would take the form **[S]**:

$$h_{\text{T0}}(f) = h_{\text{GR}}(f) \left(1 + \xi \left(\frac{f}{f_{\text{Planck}}}\right)^2\right) \tag{33}$$

In the LIGO/Virgo frequency band (10–1000 Hz), $(f/f_{\text{Planck}})^2$ is of order 10^{-85} to 10^{-81}; this term cannot be measured with present detectors.

Quantum Computer Benchmarks

T0 Quantum Error Rate

For T0 quantum computers a slightly lower error rate would result **[S]**:

$$\epsilon_{\text{gate}}^{\text{T0}} = \epsilon_{\text{gate}}^{\text{Standard}} \cdot \left(1 - \xi \frac{E_{\text{gate}}}{E_{\text{Pl}}}\right) \tag{34}$$

.8 Philosophical Implications of T0 Quantum Mechanics

Determinism vs. Quantum Randomness

The T0 theory offers a deterministic interpretation of quantum randomness:

T0 Determinism

Quantum Randomness as Effective Unpredictability: What appears as fundamental randomness in standard QM is deterministic time field dynamics in the T0 theory. These dynamics lead to practically unpredictable, but in principle determined outcomes.

$$\begin{aligned} \text{“Randomness”} = {} & \text{Deterministic} \\ & \text{Time Field Evolution} \\ & + \text{Practical} \\ & \text{Unpredictability} \end{aligned} \tag{35}$$

Interpretation of the Measurement Problem

For the measurement problem of quantum mechanics, the T0 theory proposes the following route **[S]**:

- **No Collapse:** Wave functions evolve continuously
- **Measurement Devices:** Macroscopic T0 field configurations
- **Definite Outcomes:** Deterministic time field interactions
- **Born Rule:** Expected to emerge from T0 field dynamics (not derived here)

Locality and Realism in the T0 Picture

The T0 theory aims to retain locality and realism; this requires the superdeterministic interpretation described above **[S]**:

$$\text{Locality:} \quad \text{All interactions mediated by local T0 fields} \tag{36}$$
$$\text{Realism:} \quad \text{Particles have definite properties before measurement} \tag{37}$$
$$\text{Causality:} \quad \text{No superluminal information transfer} \tag{38}$$

.9 Technological Applications

T0 Quantum Computer Architecture

Hardware Implementation

T0 quantum computers could be realized through controlled time field manipulation:

- **Time Field Modulators:** High-frequency electromagnetic fields
- **Energy Field Sensors:** Ultra-precise field measurement devices
- **Coherence Control:** Stabilization through time field feedback
- **Scalability:** Natural decoupling of neighboring qubits

Quantum Error Correction with T0

T0-specific error correction codes:

$$|\psi_{\text{kodiert}}\rangle = \sum_i c_i |i\rangle \otimes |T_{\text{field},i}\rangle \tag{39}$$

The time field could serve as a syndrome for error detection.

Precision Measurement Technology

T0-Enhanced Atomic Clocks

For atomic clocks, the T0 theory predicts the gravitational redshift of General Relativity; there is currently no prediction beyond it. In the static weak field, the mass reading $m = m_0\, e^{\Phi/c^2}$ yields the same parameters as GR, $\gamma = \beta = 1$ (Doc. 392); signatures of their own proportional to ξ in atomic clocks have not been derived so far (Doc. 129). A clock comparison across a height difference therefore primarily tests the redshift. This does not rule out deviations; a measured deviation would be a hint that still has to be located within the theory.

Note: An earlier version stated a correction $\delta f/f_0 = -\xi\,\Delta E_{\text{Transition}}/E_{\text{Pl}}$ at this point. It has been removed because it was not derived and therefore contradicted the statement "No Derived Deviation beyond the Redshift" above and Doc. 129. It had no consequences for measurements: for ytterbium it would be about 10^{-32}, and it cancels exactly when two clocks of the same atomic species are compared.

Gravitational Wave Detectors

Possible change in sensitivity through T0 field calibration **[S]**:

$$h_{\text{min}}^{\text{T0}} = h_{\text{min}}^{\text{Standard}} \cdot \left(1 - \xi\sqrt{f \cdot t_{\text{int}}}\right) \tag{40}$$

.10 Standard Model Extensions

T0-Extended Standard Model

The Standard Model is formally extended by T0 terms:

$$\mathscr{L}_{\text{SM}}^{\text{T0}} = \mathscr{L}_{\text{SM}} + \mathscr{L}_{\text{T0-Feld}} + \mathscr{L}_{\text{T0-Interaction}} \tag{41}$$

where:

$$\mathscr{L}_{\text{T0-Feld}} = \frac{\xi}{E_{\text{Pl}}^2}(\partial T(x,t))^2 \tag{42}$$

$$\mathscr{L}_{\text{T0-Interaction}} = \xi \sum_i g_i \bar{\psi}_i \gamma^\mu \partial_\mu T(x,t) \psi_i \tag{43}$$

Hierarchy Problem: T0 Approach

For the hierarchy problem, the following relation can be formed from the T0 structure:

$$\frac{M_{\text{Planck}}}{M_{\text{EW}}} = \frac{1}{\sqrt{\xi}} \approx \frac{1}{\sqrt{1.33 \times 10^{-4}}} \approx 87 \tag{44}$$

This relation does not hold: $1/\sqrt{\xi} = 86.6$ is calculated correctly, but the measured value is $E_P/v = 5 \times 10^{16}$; the relation does not reproduce the ratio of the Planck and electroweak scales.

.11 Critical Evaluation and Limitations

Experimental Challenges

The experimental verification of the T0 theory requires:

- **Ultra-High Precision**: Measurements at the 10^{-22}–10^{-31} level
- **New Technologies**: T0 field-specific measurement devices
- **Long-Term Stability**: Consistent measurements over years
- **Systematic Control**: Elimination of all other effects

Philosophical Implications

The T0 theory raises basic philosophical questions:

- **Free Will**: Is determinism compatible with human freedom of decision?
- **Epistemology**: How can we fully recognize the T0 reality?
- **Reductionism**: Are all phenomena reducible to T0 fields?
- **Emergence**: What role do emergent properties play?

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